

INTRODUCTION

Water is essential to life on our planet, and its careful management is integral to maintaining ecological equilibrium and ensuring its well-being for all living beings. This project examines water consumption and cost data to gain insight into patterns that could improve usage while decreasing costs for various stakeholders such as residents, businesses, and government agencies. The dataset contains the following columns-

1. **Development Name:** The name of the development or property where the meter is located.
2. **Borough:** The borough where the meter is located (Bronx, Brooklyn, Manhattan, Queens, or Staten Island).
3. **Account Name:** The name of the account holder or property owner.
4. **Location:** The address where the meter is located.
5. **Meter AMR:** Indicates whether the meter is capable of automated reading (AMR) (Yes or No).
6. **Meter Scope:** Indicates whether the meter measures water consumption for an individual unit or a master meter for the entire building (Individual or Master).
7. **TDS #:** The property's three-digit Tax Department Section (TDS) code.
8. **EDP:** The property's eight-digit Electric Data Processing (EDP) account number.
9. **RC Code:** The Revenue Class (RC) code for the account.
10. **Funding Source:** The funding source for the account (such as Public or Private).
11. **AMP #:** The account management program (AMP) number for the account.
12. **Vendor Name:** The name of the vendor that services the meter.
13. **UMIS BILL ID:** The Unique Management Information System (UMIS) bill identification number for the account.
14. **Revenue Month:** The month for which the bill was issued.
15. **Service Start Date:** The start date of the billing period.
16. **Service End Date:** The end date of the billing period.
17. **days:** The number of days in the billing period.
18. **Meter Number:** The identification number of the meter.
19. **Estimated:** Indicates whether the consumption for the billing period was estimated or actual (1 for estimated, 0 for actual).
20. **Current Charges:** The total charges for the billing period.
21. **Rate Class:** The rate class code for the account.
22. **Bill Analyzed:** Indicates whether the bill was analyzed or unaudited (Analyzed or Unaudited).
23. **Consumption (HCF):** The consumption in hundred cubic feet (HCF) for the billing period.
24. **Water&Sewer Charges:** The water and sewer services charges for the billing period.
25. **Other charges:** These include late fees or penalties for the billing period.

Source: <https://data.cityofnewyork.us/api/views/66be-66yr/rows.csv?accessType=DOWNLOAD>

Dataset Glimpse :

HEAD :

Development_Name	BAY VIEW
Borough	QUEENS
Account_Name	BAY VIEW
Location	BLD 25 - Community Center
Meter_AMR	NONE
Meter_Scope	Community Center
TDS	92.0
EDP	670
RC_Code	Q009100
Funding_Source	MIXED FINANCE/LLC1
AMP	NY005020920P
Vendor_Name	NEW YORK CITY WATER BOARD
UMIS_BILL_ID	8870760
Revenue_Month	2020-04
Service_Start_Date	12/23/2019
Service_End_Date	04/23/2020
Days	34.0

Through this analysis, there are vital questions that must be answered:

1. Which developments have the highest and lowest water consumption?
2. How is water usage distributed between different boroughs?
3. Which accounts have the highest and lowest current charges?
4. What percentage of meters possess Automatic Meter Reading (AMR) capability?
5. How does the billing period vary across various rate classes?
6. Do estimated and actual water consumption differ in the top 20 developments?
7. How are water usage patterns linked with revenue month and borough levels?
8. Average Days in Billing Period by Rate Class
9. Actual vs. Estimated Consumption (HCF) by Development Name (Top 20)
10. Water Consumption (HCF) by Revenue Month and Borough

We employed a combination of data processing, visualization, and machine learning techniques to address these questions. The dataset we utilized for this analysis includes information on water consumption, charges, and relevant details from 2013-2023 from 2013 through February. To clean and process this dataset properly - having by converting columns to appropriate data types while handling missing values properly and encoding categorical variables - encoding categorical variables was also completed.

We created various visualizations, such as bar charts, pie charts, and heat maps, to explore multiple aspects of our data. These visuals allow for detailed insights into water consumption patterns by development type, borough name, AMR capability of meters, and other factors.

In additionally to visuals, we have used three regression models for prediction:

1. Current charges are based on consumption and Billing information with random forest regressor.

2. Revenue estimation using Gradient Boosting regressor for one month's consumption, rate class, and additional charges.

By employing these models, we aim to equip stakeholders with beneficial data-driven decisions to optimize water usage while lowering associated costs.

This project analyses water consumption and cost data to address pertinent queries and offer insights to improve water management practices. By employing data processing, visualization, and machine learning techniques, we explored various aspects of the data and produced predictions to assist stakeholders with making informed decisions for efficient resource management.

Exploratory Data Analysis

Data Clean-up

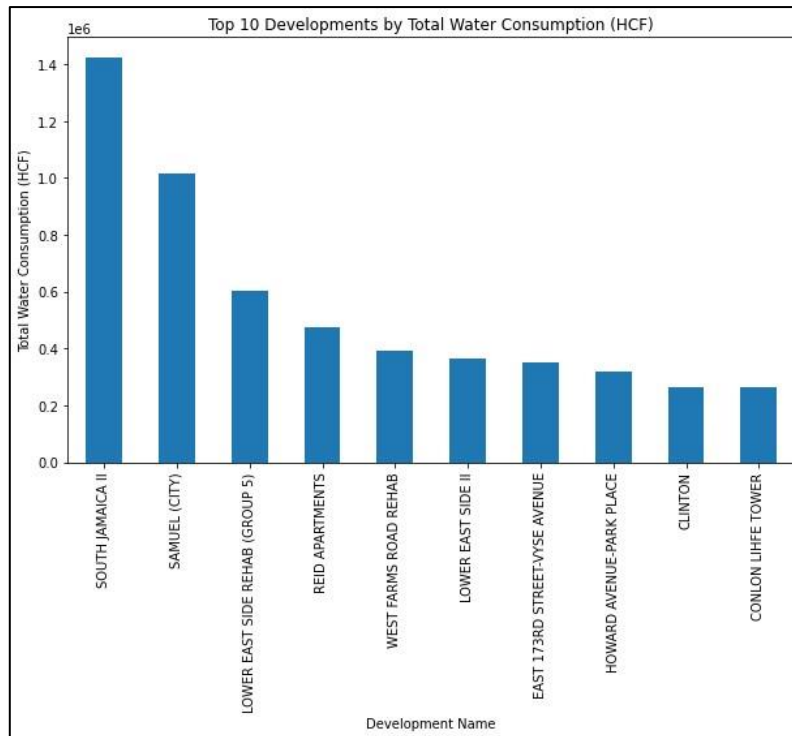
Before the analysis, we performed data pre-processing and clean-up to ensure our data is accurate and consistent. The following steps were undertaken:

- Conversion of 'Service_Start_Date' and 'Service_End_Date' columns to DateTime objects.
- Filling missing values in categorical columns with their mode.
- Filling missing values in numeric columns with their median.
- Conversion of 'TDS' and 'Days' columns to integers.

Data Visualizations

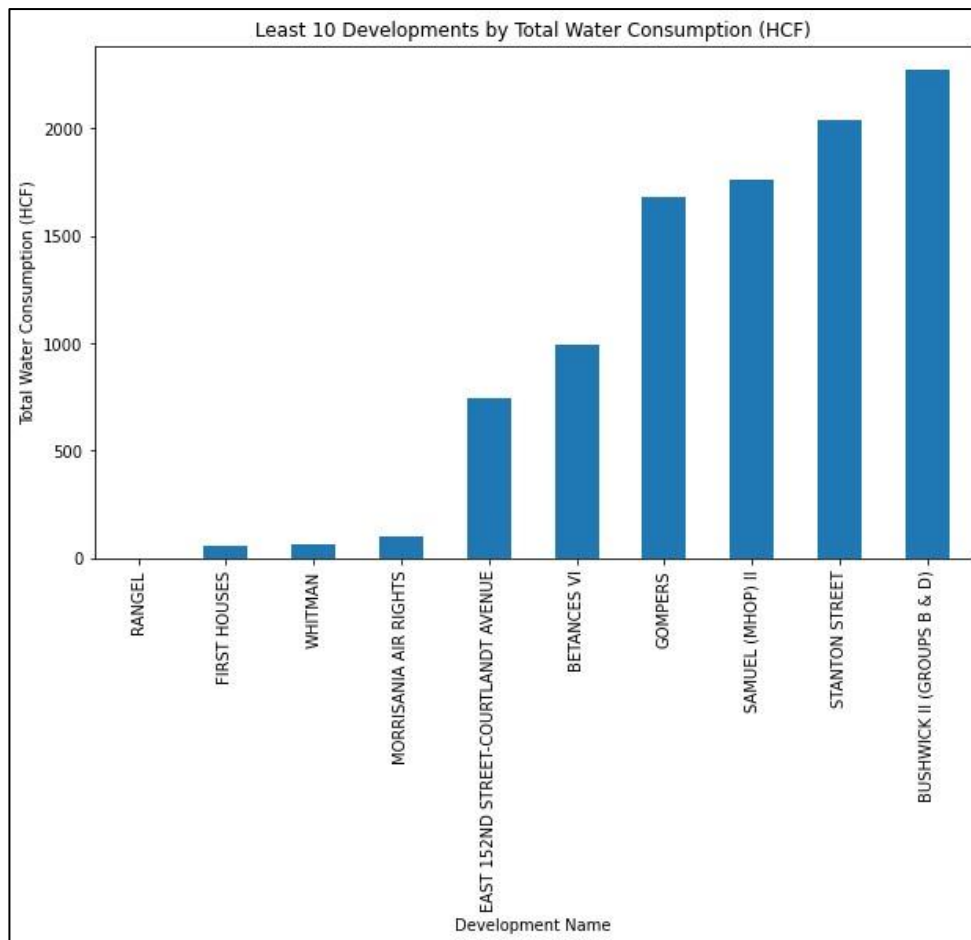
1. Top 10 Developments by Total Water Consumption (HCF)

The bar chart displays the top ten developments with the highest water usage among all dataset developments in terms of hundred cubic feet per day (HCF), providing insight into areas requiring intervention, optimization, or further examination. From our analysis, it can be seen that "South Jamaica II" had the highest water consumption, with 1,424,036 HCF consumed, followed by "Samuel (City)," which consumed 1,016,605 HCF. All remaining developments had significantly lower consumption rates, with "Lower East Side Rehab (Group 5)" accounting for 603,806 HCF, while "Conlon Lihfe Tower" consumed 263,840 HCF of water consumption. This graph allows us to identify which developments drink the most water, providing insight into targeted management or conservation efforts that could benefit from implementation in these top 10 developments. Stakeholders may implement changes that can significantly affect water consumption and enhance resource management.



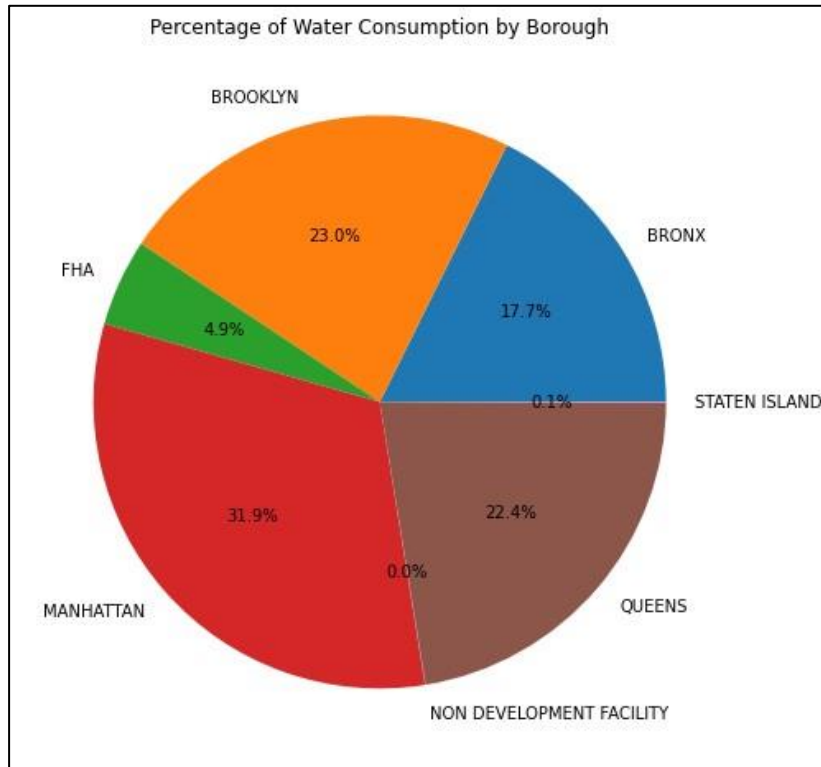
2. Least 10 Developments by Total Water Consumption (HCF)

This bar chart displays the top ten developments with the least total water usage per 100 cubic feet from our dataset. This visualization helps identify developments with minimal use which may provide insights into potential best practices, data anomalies, or areas requiring further investigation. Our data analysis shows that "Rangel" had the lowest water consumption with zero HCF consumption. This may indicate a data anomaly or that this development wasn't operational during our analysis period. Second was First Houses at 54 HCF, followed by "Whitman at 62. Lastly was Bushwick II Groups B & D), at 2,270 HCF consumption. This graph can assist us in understanding which developments consume the minor water consumption, providing an opportunity to identify best practices in water management and discover any discrepancies with data, such as missing or incorrect entries. By studying the least-consuming developments, stakeholders may gain valuable insight into their water management strategies that they could apply across other areas to enhance overall resource management and increase water resource productivity.



3. *Percentage of Water Consumption by Borough*

The pie chart displays the distribution of HCF water consumption across each borough in our dataset. This visualization helps us better understand which consumes the highest and lowest amounts. From this pie chart, we can see that Manhattan had the highest water consumption at 31.9% of total consumption; Brooklyn and Queens followed closely behind with 26.0% each; Bronx took 17.7% while Staten Island, FHAs, and non-Development Facilities each represented significantly reduced percentages ranging between 0.1%, 4.9% and 0.03% of consumption respectively. Using this visualization, stakeholders can quickly pinpoint areas with high water consumption to facilitate water resource management efforts and prioritize them accordingly. Furthermore, this gives an excellent snapshot of water consumption across boroughs for planning and decision-making purposes.



4. Top 10 Accounts by Total Current Charges

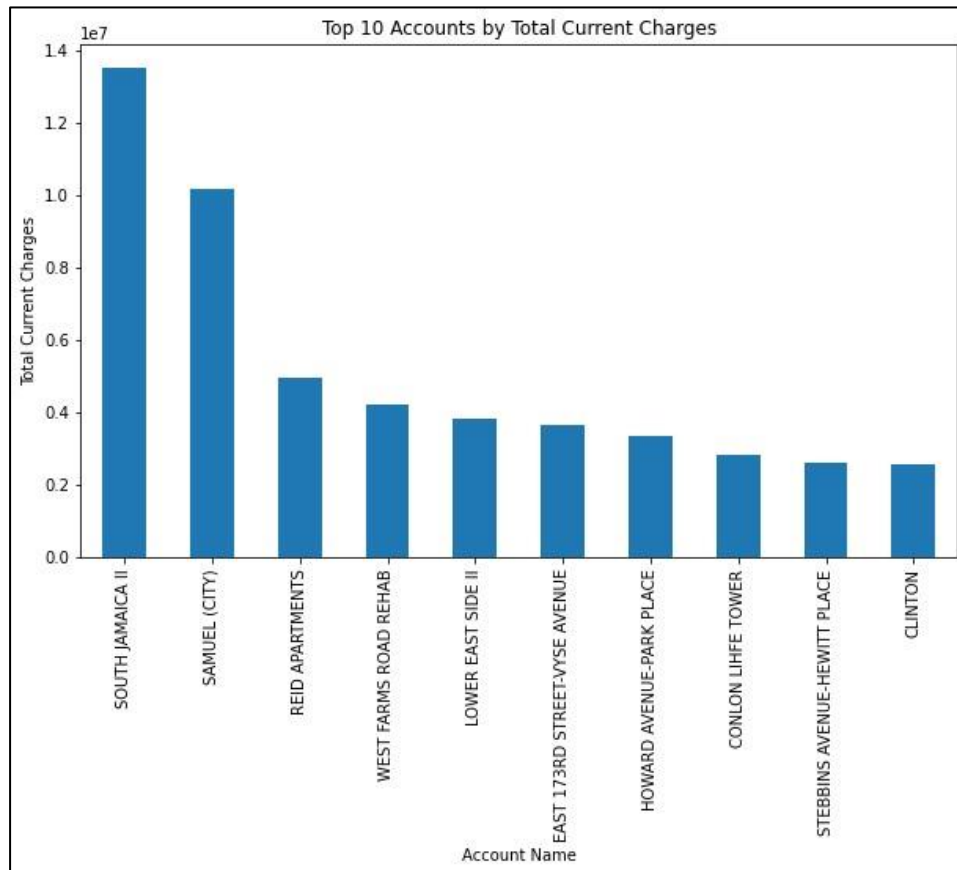
This bar chart shows which accounts have the highest charges across a dataset. This visualization makes it easier for stakeholders to monitor and manage expenses by quickly highlighting high-cost accounts.

Here is a list of the Top Ten Accounts with their current charges:

1. South Jamaica II: \$13,500,505.12
2. Samuel (City): \$10,174,790.21
3. Reid Apartments: \$4,928,621.03
4. West Farms Road Rehab: \$4,196,868.77
5. Lower East Side II: \$3,825,087.31
6. East 173rd Street-Vyse Avenue: \$3,633,200.68
7. Howard Avenue-Park Place: \$3,353,117.68
8. Conlon Lihfe Tower: \$2,805,668.68
9. Clinton: \$2,545,281.65
10. Stebbins Avenue-Hewitt Place: \$2,607,143.45

South Jamaica II tops the chart, with current charges totaling an impressive \$13,505,505.12, Samuel (City) at \$10,174,790.21, and Reid Apartments at \$4,928,621.03, respectively. All other accounts in the top ten have significantly lower current charges that range between

\$2545,281.65 (Clinton) to \$4196,868.77 (West Farms Road Rehab). Information like this is valuable in aiding stakeholders to quickly pinpoint accounts with high expenses and uncover any underlying reasons contributing to excessive expenditures. Furthermore, such data can assist decision-making processes relating to budget allocation, cost reduction strategies, and resource management.



5. Least 10 Accounts by Total Current Charges

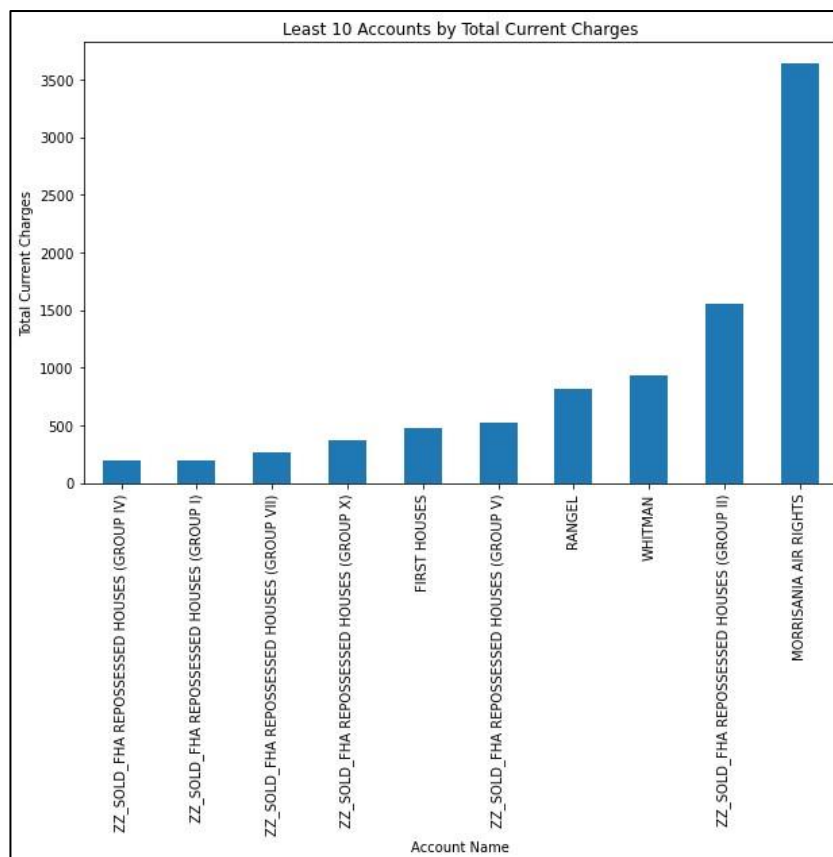
This bar chart displays accounts with the lowest total current charges within a dataset. This visualization helps stakeholders gain visibility of accounts with minimal expenses that might help explain lower expenses overall.

Here is the following list of accounts with their associated current charges:

1. ZZ_SOLD_FHA Repossessed Houses (Group IV): \$191.81
2. ZZ_SOLD_FHA Repossessed Houses (Group I): \$191.81
3. ZZ_SOLD_FHA Repossessed Houses (Group VII): \$262.11
4. ZZ_SOLD_FHA Repossessed Houses (Group X): \$371.71
5. First Houses: \$474.13
6. ZZ_SOLD_FHA Repossessed Houses (Group V): \$524.22

7. Rangel: \$816.03
8. Whitman: \$928.30
9. ZZ_SOLD_FHA Repossessed Houses (Group II): \$1,550.81
10. Morrisania Air Rights: \$3,641.05

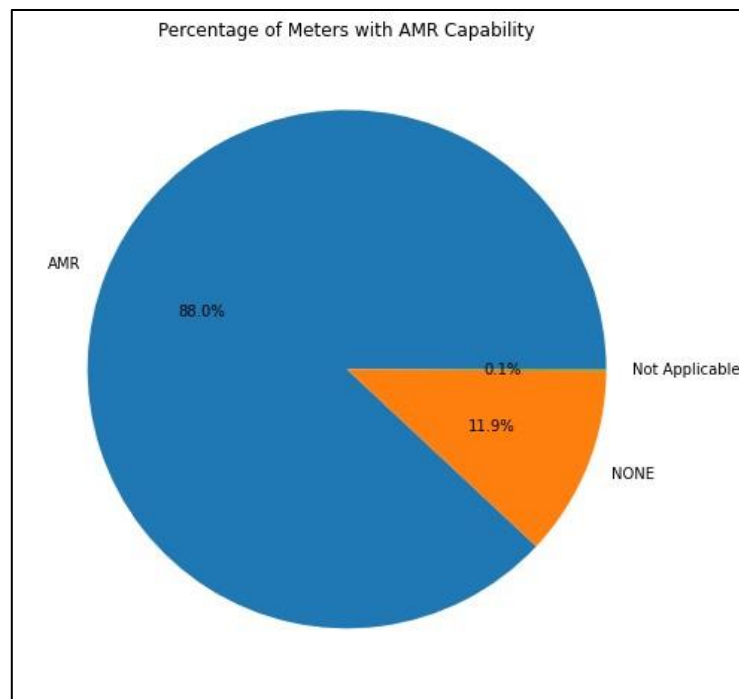
This chart indicates that accounts with the lowest current charges are mostly those marked "ZZ_SOLD_FHA Repossessed Houses," with charges between \$191.81 to \$1,550.81. First Houses, Rangel Whitman Air Rights accounts, and Morrisania Air Rights also generally have relatively reasonable monthly charges ranging from \$474.13 (First Houses) to \$3641.05 (Morrisania Air Rights). Understanding what factors contribute to lower account expenses can offer valuable insights for stakeholders. They can use this knowledge to investigate cost-cutting measures or devise plans to reduce costs in other accounts with higher expenses.



6. Percentage of Meters with AMR Capability

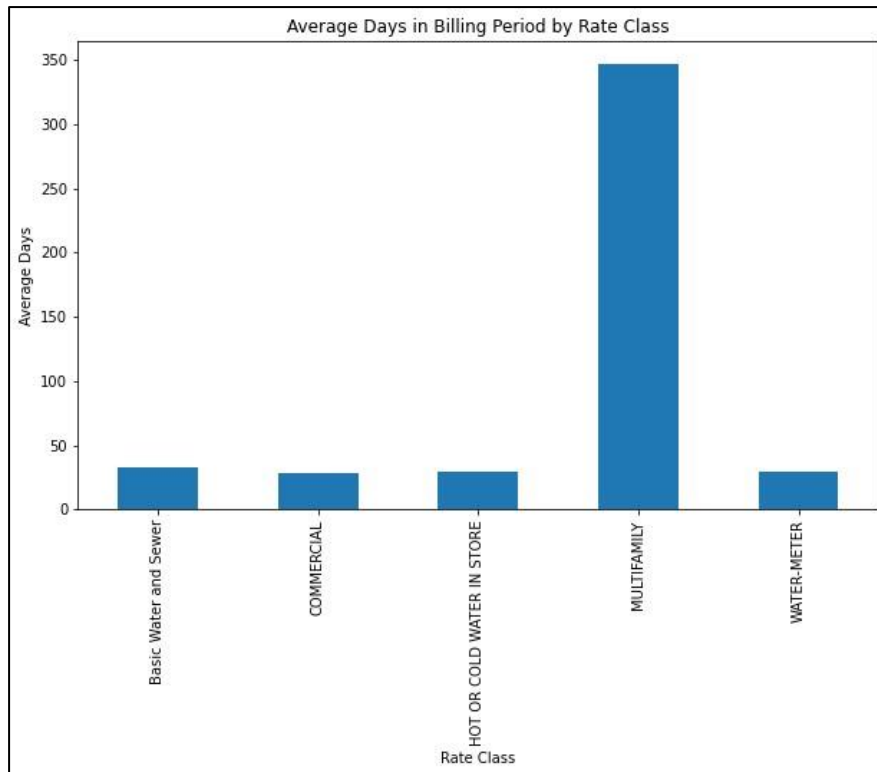
This pie chart displays the percentage of meters equipped with Automatic Meter Reading (AMR) capabilities. AMR technology enables remote data collection from water meters, decreasing manual meter reading costs while improving billing accuracy and precision. This chart indicates that 88.1% of meters use AMR technology, an impressive statistic that speaks volumes for how we have adopted modernity in water consumption monitoring. AMR's widespread adoption can contribute to more accurate billing, lower labor costs, and enhanced customer service. But there remains room for growth as 11.9% of meters do not yet support AMR technology, suggesting there could be room for upgrade efforts to boost data collection efficiency and ensure accurate collections processes. At last, 0.08% of meters have been

classified as "Not Applicable," potentially meaning they no longer run or no longer need servicing. The investigation may be required to understand why these meters were classified this way and develop the appropriate action for these cases.



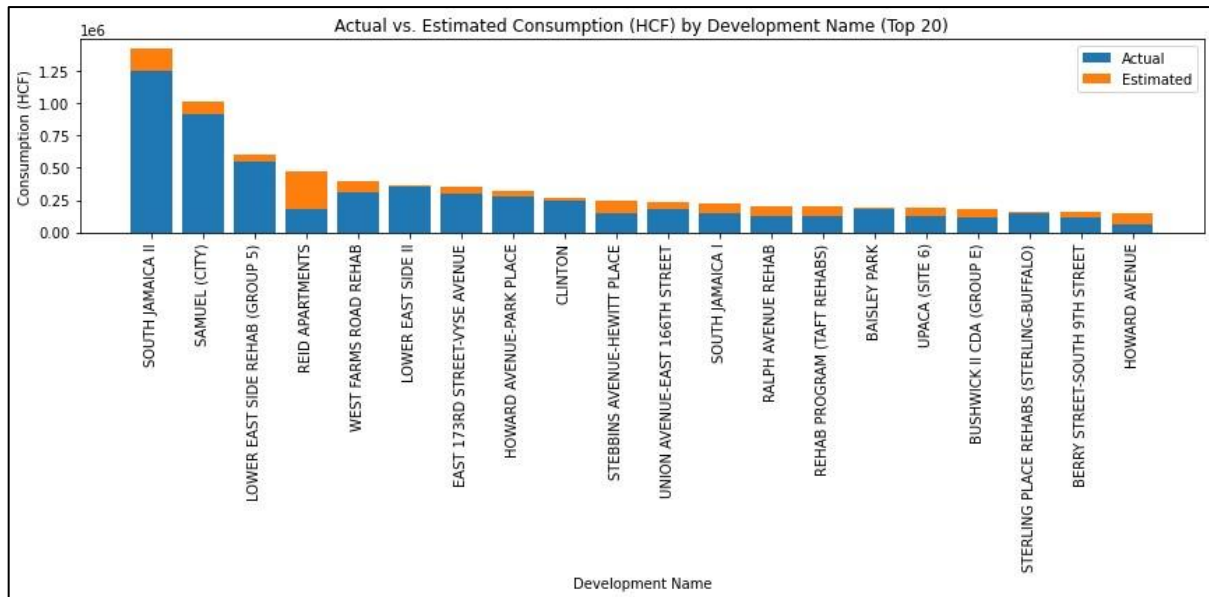
7. Average Days in Billing Period by Rate Class

This bar chart depicts the average length of billing periods by rate class. Each rate class provides a pricing structure for water and sewer services based on property or usage characteristics. The chart indicates that the MULTIFAMILY rate class has an unusually long billing cycle (347 days). This may be related to multifamily properties typically comprising multiple units sharing water and sewer infrastructure requiring separate billing cycles than other properties. Other rate classes, including WATER-METER, HOT OR COLD WATER IN STORE, COMMERCIAL and Basic Water and Sewer properties, tend to have shorter billing periods between 28.56-33.18 days for accurate billing cycles that ensure more timely collection. Knowledge of billing periods across rate classes can assist utilities and property owners in better-controlling water consumption and billing processes while helping identify discrepancies or inefficiencies within their billing system that require further examination or enhancement.



8. Actual vs. Estimated Consumption (HCF) by Development Name (Top 20)

This stacked bar chart compares actual to estimated (HCF) water usage data across 20 developments that comprise the Top 20, providing insight into whether calculated consumption estimates are accurate and any possible discrepancies with actual usage data. Each development in the Top 20 list displays two bars side-by-side; one shows actual water consumption while the other represents estimated usage. Comparing them reveals whether estimates were accurate and any significant disparity between actual and estimated usage figures. By studying this chart, we can quickly and efficiently identify developments where estimates are precise relatively or indicate significant discrepancies that could suggest estimation processes or infrastructure issues. Understanding such differences enables utilities and property managers to take appropriate actions to enhance water consumption monitoring, billing accuracy, and overall water management practices. For an improved interpretation, it would be helpful to provide actual and estimated consumption values (in HCF) for each development to more easily quantify any discrepancies between estimated and basic consumption levels. This data would enable more precise evaluation and comprehension.



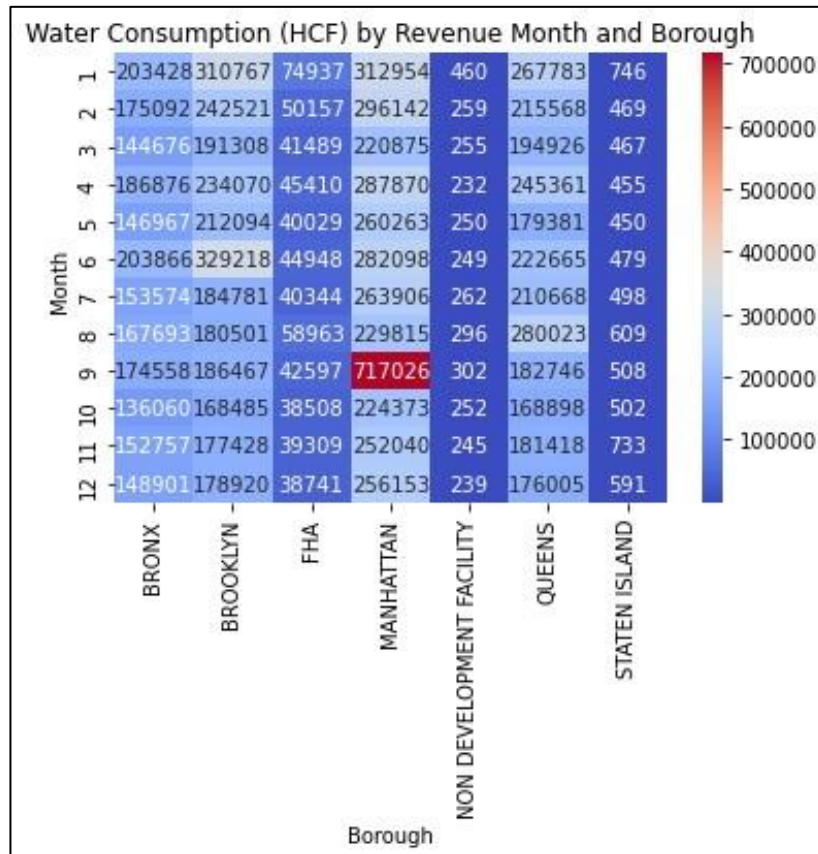
9. Water Consumption (HCF) by Revenue Month and Borough

The heatmap illustrates water consumption across New York City boroughs for each month during 2012 by using color intensity to show levels of HCF consumption; darker shades indicate greater consumption levels.

By studying a heatmap, we can gain valuable insights and gain awareness of these trends:

1. Manhattan consistently experiences high water consumption throughout the year. Water use reached 717,026 HCF in September alone, necessitating further investigation to understand the underlying factors causing such high usage.
2. Manhattan, Brooklyn, and Queens typically experience relatively high water consumption levels throughout the year, though their consumption does not show significant swings.
3. Compared with other boroughs, The Bronx exhibits lower water consumption levels - though some months, such as January and June, have slightly increased usage levels.
4. Staten Island consistently consumes less water in all five boroughs, and consumption numbers remain consistently low overall the months.
5. FHA and Non-Development Facility categories have consistently low water usage levels each year, as do Non-Development Facility properties or facilities in general. Neither FHA nor Non-Development Facilities fall into geographical boroughs like any other properties would do; instead, they represent specific properties or facilities.

This heatmap offers an interactive visual depiction of water consumption across boroughs and categories over an annual timescale, providing stakeholders with an overview of trends, potential issues, or areas for improvement in terms of both management and consumption of the resource.



Predictive Models

This project uses three predictive models to investigate various aspects of water consumption data, using methods such as RandomForestRegressor, DecisionTreeRegressor and LabelEncoder combined with RandomForestRegressor to predict current charges, estimated charges, and revenue, respectively. We selected these methods based on their ability to address complex and nonlinear relationships found within water consumption records.

1. Random Forest Regression for Predicting Current Charges:

Random Forest Regression was chosen for predicting current charges because it is an ensemble method combining multiple decision trees. This model can effectively handle complex relationships and nonlinear patterns within data. By aggregating results from numerous decision trees, Random Forest Regression can reduce overfitting and improve prediction

accuracy. We trained this model on consumption (HCF) and Billing Information data and found it could accurately capture patterns while making accurate predictions against test data sets.

Splitting dataset into Training and Testing :

Predict_current_charges: The model is trained using the features 'Consumption_(HCF)', 'Other_Charges', 'TDS', 'Water&Sewer_Charges' and predicts the target variable 'Current_Charges'. The training and testing sets are created by splitting X and y with a test size of 0.2 (20% of the data) and a random state of 42.

```
In [5]: ► Predict_current_charges(df)

Out[5]: (    predicted_Current_charges
0          43.150000
1       1478.563884
2        151.499101
3       447.091800
4         65.166799
...
10058      2681.433827
10059      1548.854975
10060      7619.218250
10061         89.631928
10062         65.166799

[10063 rows x 1 columns],
{'Accuracy': 0.8152333478350529})
```

2. Gradient boosting Regression to predict revenue:

Last, we used Gradient Boosting Regression to predict revenue. LabelEncoder served to convert categorical Rate_Class variables to numerical values so Gradient boosting Regression could process them effectively and predict revenue effectively on test data. The trained model performed satisfactorily against this task on actual test data sets.

In summary, the chosen predictive models proved appropriate to their tasks as they can effectively handle complex datasets while simultaneously recognizing hidden patterns. Gradient Boosting effectively forecasted revenue. These model provides valuable insight into the factors affecting water consumption and charges, which may influence policy and resource allocation decisions. Furthermore, by interpreting their results, we gained greater clarity into relationships among consumption charges revenue for more informed decision-making regarding usage and management.

Splitting dataset into Training and Testing :

Predict_revenue: The model is trained using the features 'Consumption_(HCF)', 'Other_Charges', 'Rate_Class_Encoded', 'TDS', 'Service_Start_Date', 'Service_End_Date' and predicts the target variable 'Current_Charges'. Similarly, the training and testing sets are created with a test size of 0.2.

```
In [22]: ► Predict_revenue(df)

Out[22]: (      predicted_Revenue
0          43.150000
1        1480.094487
2         152.048958
3         447.356386
4          51.593920
...
10058        2718.187829
10059        1551.991412
10060        7632.746037
10061          89.627646
10062          51.593920

[10063 rows x 1 columns],
{'Accuracy': 0.7981789745613128})
```

Interpretive and Conclusions

As part of this project, we analyzed water consumption data to answer two main questions about its influencers and usage:

(1) Which factors impact consumption/charges patterns/trends? (2) Can this information help make informed decisions for usage/management practices?

We gained critical insight into patterns/trends regarding consumption/charges/revenue streams through various data visualization techniques, exploratory analysis techniques, and predictive models. Our analysis showed that water consumption can vary substantially among different developments and boroughs. We identified top developments by total consumption and current charges as well as low users; many meters had AMR capability, providing more accurate meter readings and billing. Our analysis used three distinct predictive models to delve deeper into the data: RandomForestRegressor for current charges; Gradient Boosting as revenue predictions models. Both the models proved capable of accurately capturing patterns within test data and making reliable forecasts about it. Our findings and analyses show that water consumption and charges depend heavily on development size, location, meter type, and rate class. With these insights, policymakers, utility providers, and other stakeholders may make more informed water management and usage decisions.

Suggestions

1. Target high-consumption developments and accounts: Concentrating interventions and awareness programs at top water-consuming developments and accounts can bring more significant water savings and better management practices.
2. Increase use of AMR meters: Adopting AMR-enabled meters will facilitate more precise billing and monitoring, helping users make more informed choices regarding their water consumption.
3. Understand the Impact of Rate Classes on Revenue: Understanding how different rate classes impact revenue can help create policies and pricing structures supporting water conservation while remaining financially sound.
4. Keep tabs on estimated charges: By monitoring how often they appear in their billing systems, utility providers can identify areas where meter reading and billing accuracy could be enhanced.
5. Regularly review and adjust water management strategies: With water consumption patterns and costs evolving over time, strategies for water management must be periodically assessed against new data to stay ahead of any shifts or insights that come to light.

Overall, this project's analysis provided invaluable insights into the factors affecting water consumption and charges. By drawing upon this evidence-based knowledge, stakeholders could use them to make more informed decisions about their usage or management - ultimately contributing to sustainable use of resources such as water.

Reference

1. Agarwal, M. (2023, February 7). Pythonic data cleaning with pandas and NumPy. Real Python. <https://realpython.com/python-data-cleaning-numpy-pandas/>
2. Shatby, S. E. (2022, May 18). How to build a predictive model in python?. 365 Data Science. <https://365datascience.com/tutorials/python-tutorials/predictive-modelpython/>